

demonstrate that MAX binding stabilized c-MYC with the reduction in the RMSF profile, increased radius of gyration, and a decrease in the mean backbone root-mean-square displacement from 10.55 to 2.66 Å. The c-MYC:MAX interface remained stable throughout the simulations, supported by 437 persistent atomic contacts and possesses mainly two interchain salt bridges: LYS256(MAX)-GLU957(MYC) (with ~90% occupancy) and ARG970(MYC)-GLU275(MAX) (with ~70% occupancy). To identify druggable regions at the c-MYC:MAX interface, we integrated three complementary approaches: MDpocket cavity detection, MM-GBSA per-residue energy decomposition, and salt bridge persistence analysis. MDpocket identified six sub-pockets at the interface, and pocket SP2 emerged as the primary druggable candidate based on its consistent detection frequency and its overlap with hotspot residues identified by the other two methods. Together, these three methods provide a residue-level description of the c-MYC:MAX binding interface, converging on eight specific residues as candidates for structure-based inhibitor design: ARG913, ARG914, GLU916, and ARG970 in MYC, and ARG239, GLU248, ARG254, and GLU275 in MAX.

Introduction

The MYC family of proto-oncogenes encodes a pleiotropic, basic helix-loop-helix transcription factor that coordinates a vast transcriptional network. Functioning as a master regulatory hub, MYC is essential for maintaining cellular homeostasis, driving cell cycle progression, and dictating metabolic reprogramming. While its tightly calibrated activity is physiologically required for development, tissue regeneration, and DNA damage repair,¹⁻⁴ genomic alterations or aberrant expression often lead to tumorigenesis. Because of its fundamental regulatory roles, the MYC sequence is highly conserved across phylogenetically diverse species.⁵ Initially discovered in the avian retrovirus MC29,^{6,7} c-MYC remains one of the earliest identified proto-oncogenes. Its enduring evolutionary conservation and central involvement in biological processes further underscore its critical significance in both

healthy and pathological cellular states. MYC proteins consist of three key regions: a trans-activation domain at the N-terminus, a central region, and the bHLHLZ domain at the C-terminus, which is characteristic of a superfamily of transcription regulators that mediate cellular responses to extracellular and intracellular signals by modulating gene expression.⁸ Such domain is structured into a basic region (for DNA binding), a helix-loop-helix (HLH) motif, and a leucine zipper (LZ). It forms a continuous alpha-helical structure that allows MYC to dimerize with MAX,^{9,10} creating a functional DNA-binding four-helix bundle.

In up to 70% of human cancers,¹ including Burkitt lymphoma, colorectal, and breast malignancies,^{7,11} MYC overexpression drives an excess of MYC-MAX complexes, fueling uncontrolled cell growth. Because this interaction is crucial for oncogenesis, modern therapeutic strategies focus directly on disrupting the MYC-MAX dimer. Primary approaches include utilizing small-molecule inhibitors to block the physical binding interface between MYC and MAX, and employing bHLHLZ decoy peptides that mimic the dimerization domain to competitively intercept individual monomers. Other approaches include the discovery of new druggable pockets on the surfaces of isolated MYC¹² and MAX,¹³ the targeting of which is currently under investigation. Additionally, researchers are developing strategies to promote the alternative MAX-MXD interaction, effectively sequestering MAX and preventing the formation of the oncogenic MYC-MAX complex.

Despite the structural challenges of targeting the intrinsically disordered MYC protein, which lacks stable binding pockets¹⁴⁻¹⁷ and presents a large, flat interaction interface, several innovative strategies have emerged to disrupt the oncogenic MYC-MAX dimer.^{1,18,19} Small-molecule inhibitors, such as 10074-G5²⁰ and 10058-F4,²¹ can overcome this lack of structural definition by binding directly to specific segments of the monomeric basic bHLHLZ domain, inducing local structural distortions that prevent hybridization with MAX. Alternatively, dominant-negative peptides like Omomyc²² capitalize on a mutated bHLHLZ sequence to competitively sequester both monomers into inactive complexes, while simultaneously displacing endogenous MYC-MAX dimers from DNA binding sites to halt transcription. Col-

lectively, these direct approaches, alongside methods that shift cellular balances towards non-oncogenic MAX-MXD complexes, provide viable clinical strategies to shut down the hyperactivated MYC signaling network driving human malignancies. Although direct MYC inhibitors have not yet been approved research continues, with promising therapeutic agents currently under development and undergoing clinical trials.

In this work, we will focus on disrupting the dimerization of MYC-MAX by employing protein to protein disruption. In order to locate potential druggable pockets, we leverage Molecular Dynamics (MD) simulations at the all-atom level, which provide high temporal resolution and a precise force field (Charmm36m) to model the behavior of MYC and its interactions with other proteins and drugs. All MD simulations in this work are initiated from the crystallized MYC-MAX-DNA ternary complex (PDB: 1NKP), with the DNA coordinates omitted to isolate the protein-protein interface from the rigid structural scaffolding of the DNA backbone, allowing us to capture the unconstrained conformational relaxation of the heterodimer. On the one hand, by analyzing the conformational dynamics of isolated MYC we aim to observe the flexible nature of the protein and quantify the effects of MAX in its structural stability upon binding. On the other hand, we will use tools like MDpocket²³ and MM-GBSA²⁴ to analyse our MD data and identify pockets or interface cavities at the MYC-MAX interface that can serve as potential drug-binding sites, enabling the development of new therapeutic strategies.

Methods

System preparation and setups

In the present work we conducted MD simulations of two systems: isolated c-MYC and the c-MYC:MAX complex, both derived from the crystal structure deposited in the RCSB Protein Data Bank under accession code 1NKP.^{25,26} The DNA chain, solvent molecules and engineered disulfide linker present in the original crystal structure were removed prior to sys-

tem preparation. Each system was fully solvated by TIP3P water molecules²⁷ in potassium chloride at a physiologically relevant concentration (0.15 M). The total number of particles in the simulation boxes was 167,564 for c-MYC alone and 175,350 for the MYC-MAX system. All inputs were generated by the CHARMM-GUI solution builder²⁸⁻³⁰ adopting the CHARMM36m force field.³¹ All bonds involving hydrogens were set to fixed lengths using the LINCS algorithm.³²

The GROMACS 2023 package was employed³³ for system minimization, equilibration and production simulations. The system was minimized with a steepest descent method until convergence to $F_{\max} \leq 1000 \text{ kJ mol}^{-1}$. After energy minimization, a time step of 2 fs was used in all subsequent simulations. The particle mesh Ewald (PME) method was used for the treatment of long-range electrostatic interactions, with a Coulomb cutoff of 1.2 nm.³⁴ Van der Waals interactions were treated with a force-switch modifier between 1.0 and 1.2 nm. Three independent replicas were generated per system, differing in equilibration lengths (500 ps, 1000 ps and 1500 ps) under the NVT ensemble. Equilibration was performed in two consecutive stages of equal length per replica (250+250 ps, 500+500 ps, and 750+750 ps for replicas 1-3, respectively), following the CHARMM-GUI default equilibration protocol.³⁰ Harmonic position restraints were applied to non-hydrogen protein atoms throughout equilibration and released in a stepwise manner: the first stage used force constants of 400.0 $\text{kJ mol}^{-1} \text{ nm}^{-2}$ (backbone) and 40.0 $\text{kJ mol}^{-1} \text{ nm}^{-2}$ (side chains), reduced to 100.0 and 20.0 $\text{kJ mol}^{-1} \text{ nm}^{-2}$, respectively, in the second stage. Production simulations were performed under the NPT ensemble using the V-rescale thermostat at 298.15 K³⁵ and the C-rescale barostat at 1.0 bar.³⁶ Each replica of both systems were simulated for 600 ns, yielding a total of 1.8 μs of production trajectories of each system. Periodic boundary conditions were applied in all three directions.

Data analysis and visualization

Backbone RMSD was calculated as

$$\rho(t) = \sqrt{\frac{1}{N} \sum_{i=1}^N w_i (x_i(t) - x_i^{\text{ref}})^2} \quad (1)$$

where $x_i(t)$ and x_i^{ref} are the coordinates of atom i at time t and in the reference structure, respectively, and N is the number of atoms in the selection. Backbone coordinates were optimally superimposed onto the reference at each frame prior to RMSD calculation, and atoms were weighted equally ($w_i = 1$). RMSD reports the overall conformational drift of the protein from the reference (crystal) structure over the trajectory.

Per-residue RMSF was calculated as

$$\phi_i = \sqrt{\langle (x_i - \langle x_i \rangle)^2 \rangle} \quad (2)$$

where $\langle x_i \rangle$ is the time-averaged position of atom i (C α) following alignment of the trajectory to a reference structure, and $\langle \cdot \rangle$ denotes the time average. RMSF was computed without mass weighting and reports the local flexibility of each residue around its mean position.

The radius of gyration was calculated as

$$R_g = \sqrt{\frac{1}{M} \sum_i m_i (r_i - r_{\text{com}})^2} \quad (3)$$

where m_i is the mass of atom i , $M = \sum_i m_i$ is the total mass of the selection, r_i is the position of atom i , and r_{com} is the center of mass of the selection. R_g is mass-weighted and reports the overall compactness of the protein, increasing upon unfolding or extension and decreasing upon compaction. Inter-chain interface contacts were defined as pairs of heavy atoms within a cutoff distance of 4.5 Å, and inter-chain salt bridges were identified using a cutoff of 4.0 Å, retaining only those present in all three replicas with a mean occupancy above

30%. Per-residue binding free energy contributions were estimated using the MM-GBSA decomposition method as implemented in `gmx_MMPBSA` 1.6.4.³⁷ Given the tendency of MM-GBSA to overestimate absolute binding energies, results have been interpreted solely in terms of relative per-residue contributions for hotspot residue ranking. Data processing and visualization were performed using NumPy 1.26.4,³⁸ SciPy 1.14.1³⁹ and Matplotlib 3.7.3.⁴⁰ Molecular visualization was carried out using VMD 2.0.⁴¹

Results and discussion

The results are organized into four sections. In section 1 we characterize the conformational dynamics of an isolated c-MYC motif, in section 2 we quantify the structural stabilization induced by MAX binding, in section 3 we analyze the persistence and composition of the MYC-MAX binding interface, and finally in section 4 we identify druggable cavities at the interface through a three-method validation pipeline combining trajectory-based pocket detection, per-residue MM-GBSA decomposition and persistent salt bridge analysis.

1. Conformational disorder of isolated c-MYC

The structural dynamics of the isolated c-MYC were characterised across three independent 600 ns production trajectories. The backbone RMSD with respect to the crystal structure remained high throughout all replicas, with a combined mean of 10.55 ± 3.30 Å, and no stable plateau being reached at any point during the simulations (Figure 1A), indicating high structural flexibility in the isolated c-MYC. Notably, individual replicas showed substantial variability in mean RMSD (Replica 1: 13.32 ± 3.53 Å; Replica 2: 6.90 ± 2.68 Å; Replica 3: 11.42 ± 3.12 Å), suggesting that isolated c-MYC explores distinct conformational ensembles across independent simulations initiated from the same starting structure, in full consistency with the behaviour of an intrinsically disordered protein (IDP).⁴²

Despite this global variability, per-residue flexibility analysis revealed a consistent pattern across all three replicas (mean RMSF: 4.75, 4.17 and 4.02 Å respectively), with pronounced flexibility at the N-terminal region (residues 897-906, RMSF > 6 Å) and C-terminal tail (residues 980-984, RMSF > 6 Å). This suggests that while c-MYC explores different global conformations, the local flexibility profile of individual residues is an intrinsic property of the sequence, independent of starting conditions. Taken together, these results indicate that isolated c-MYC may lack the stable three-dimensional folded structure, motivating the characterisation of the MYC-MAX complex as the relevant structural state for interfacial

pocket identification and Protein-Protein Interaction (PPI) inhibitor design.

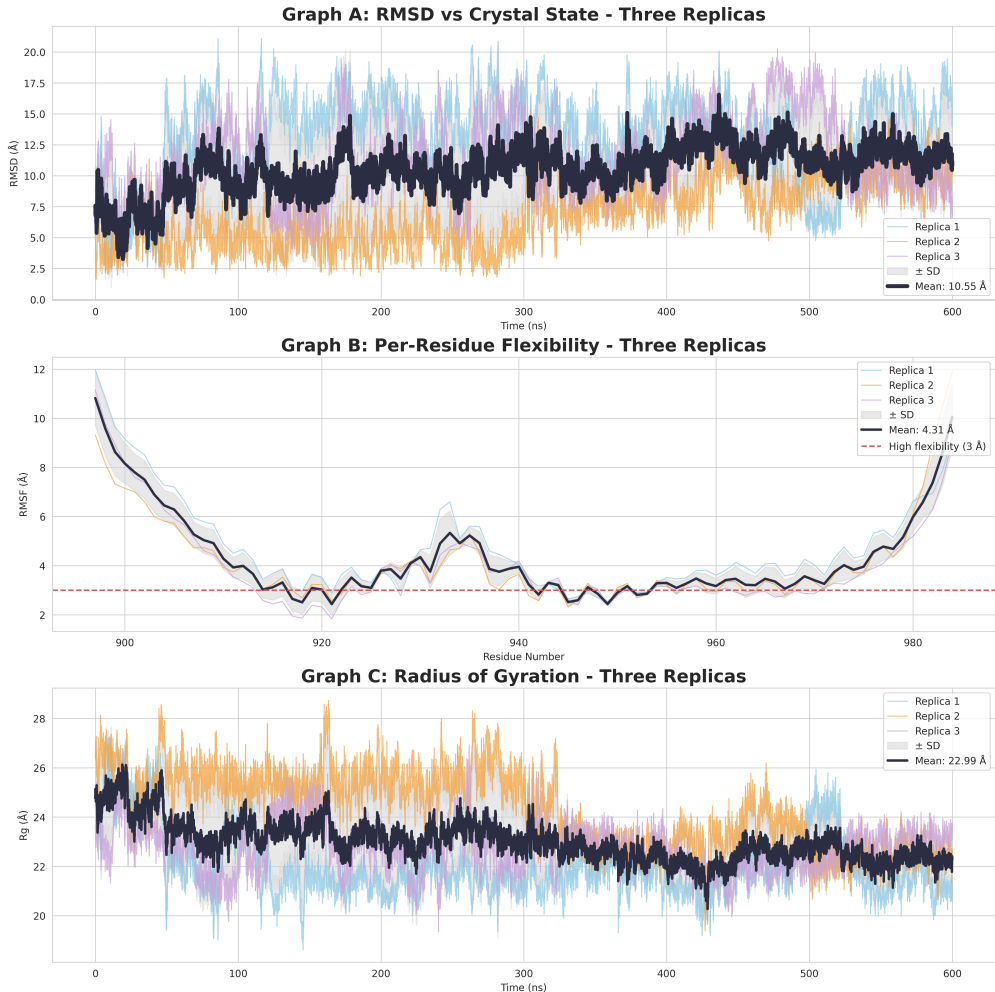


Figure 1: Structural dynamics of isolated c-MYC across three independent replicas. Graph A: backbone RMSD with respect to the crystal structure. Graph B: per-residue RMSF, with the 3 Å high-flexibility threshold indicated by a dashed red line. Graph C: radius of gyration. Shaded areas represent ± 1 standard deviation across the mean of the replicas.

2. MAX binding stabilises c-MYC conformation

To assess the structural impact of MAX on c-MYC dynamics, backbone RMSD, per-residue RMSF and radius of gyration were compared between isolated c-MYC and the c-MYC:MAX complex across three independent replicas (Figure 2).

MAX binding resulted in a dramatic reduction in backbone RMSD, from a global mean of 10.55 ± 3.30 Å in isolated c-MYC to 2.66 ± 0.31 Å in the complex ($\Delta\text{RMSD} = -7.89$ Å).

Critically, inter-replica variability was also greatly reduced in the complex (Replica 1: 2.95 ± 1.10 Å; Replica 2: 2.69 ± 0.51 Å; Replica 3: 2.34 ± 0.87 Å), in contrast to the wide spread observed for isolated c-MYC (6.90-13.32 Å across replicas). This convergence indicates that MAX binding constrains c-MYC to a well-defined conformational state regardless of initial conditions. Per-residue flexibility analysis corroborated these findings. Mean RMSF decreased from 4.31 ± 0.38 Å in isolated c-MYC to 1.80 ± 0.19 Å in the complex ($\Delta\text{RMSF} = -2.51$ Å), with the majority of residues falling below the 3 Å high-flexibility threshold upon MAX binding. The N-terminal and C-terminal regions, which exhibited RMSF values exceeding 6 Å in isolated c-MYC, showed markedly reduced fluctuations in the complex, indicating that MAX stabilises the entire bHLHLZ domain. Analysis of the radius of gyration revealed that c-MYC adopts a more extended conformation in the complex (26.61 ± 0.087 Å) compared to the isolated protein (22.99 ± 1.19 Å, $\Delta\text{Rg} = +3.62$ Å). This increase is consistent with the known behaviour of intrinsically disordered proteins, which tend to collapse into compact disordered globules in isolation and adopt extended, organised conformations upon binding to a structured partner.⁴²

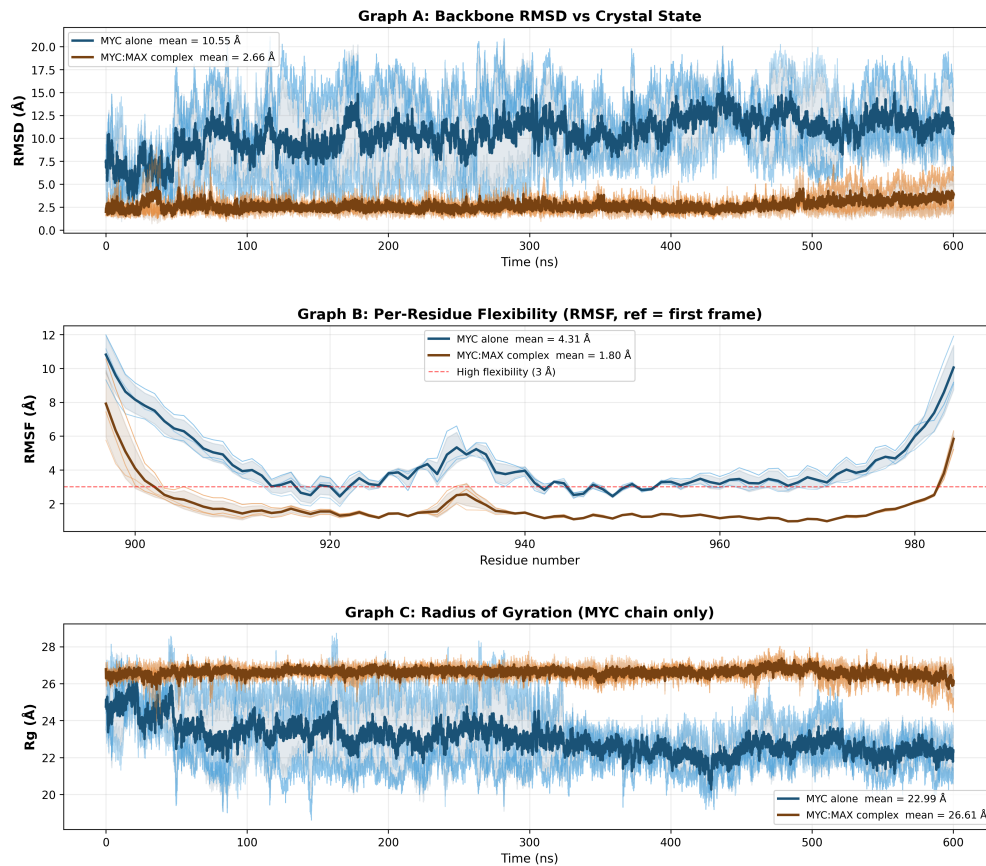


Figure 2: Comparative structural dynamics of isolated c-MYC and the c-MYC:MAX complex. Graph A: backbone RMSD with respect to the crystal structure. Graph B: per-residue RMSF, with the 3 Å high-flexibility threshold indicated by a dashed red line. Graph C: radius of gyration (R_g) of the MYC chain. Solid lines represent the mean across three replicas; shaded areas represent ± 1 standard deviation.

3. The MYC-MAX binding interface is stable and persistent

To characterise the MYC-MAX binding interface, inter-chain atomic contacts (cutoff 4.5 Å) were monitored throughout the three production trajectories. The interface maintained a mean of 437 contacts across all replicas (Replica 1: 438; Replica 2: 432; Replica 3: 440), with no temporal drift observed over the 600 ns analysis window (Figure 3). The high reproducibility across replicas confirms that the MYC-MAX complex remains stably assembled throughout the simulations and that the interface characterised here is not dependent of initial conditions.

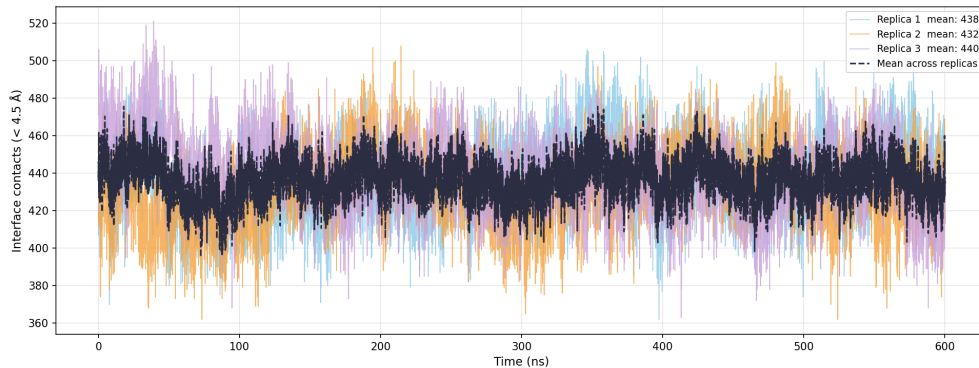


Figure 3: Mean MYC-MAX interface contacts per replica. Inter-chain atomic contacts were counted using a 4.5 Å cutoff.

Analysis of persistent inter-chain salt bridges (cutoff 4.0 Å, occupancy threshold 30%) identified two conserved electrostatic interactions present across all three replicas (Figure 4): LYS256(MAX)-GLU957(MYC), with a mean occupancy of $\sim 90\%$, and ARG970(MYC)-GLU275(MAX), with a mean occupancy of $\sim 70\%$. Both interactions were maintained well above the 30% threshold in all replicas, with small inter-replica standard deviations, confirming their reproducibility. The high occupancy of LYS256-GLU957 in particular suggests this interaction acts as a primary electrostatic anchor at the MYC-MAX interface. The role of ARG970(MYC) and GLU275(MAX) as persistent structural anchors is further supported by their recurrence as an energetic hotspot^{43,44} in the multi-method validation pipeline analysis presented in Section 4 .

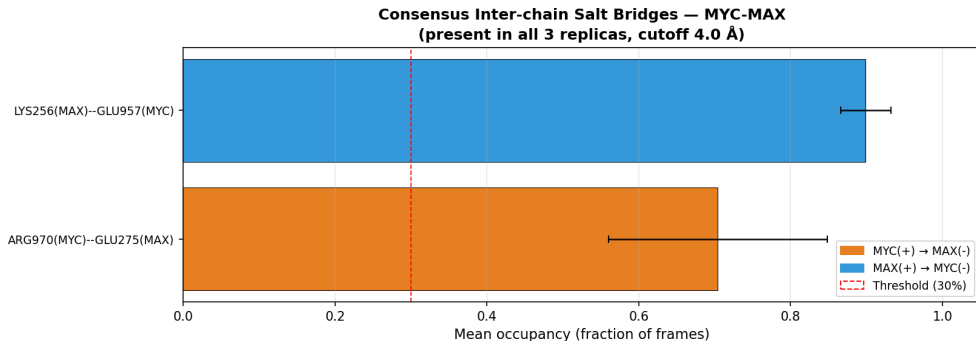


Figure 4: Consensus inter-chain salt bridges in the MYC-MAX complex, present in all 3 replicas. Cutoff was set to be 4.0 Å. Only interactions present in all three replicas with mean occupancy above 30% are shown. Error bars represent ± 1 standard deviation across replicas. The orange and blue bars indicate the MYC(+)→MAX(-) and MAX(+)→MYC(-) interactions, respectively. The direction of the interaction depends on the residues from each protein (donor (Arg, Lys, His) → acceptor (Asp, Glu)).

4. Druggable pocket characterisation at the MYC-MAX interface

To identify and validate druggable cavities at the MYC-MAX interface, a three-method validation pipeline was applied: trajectory-based pocket detection with MDpocket²³ for searching potential drug pocket on the surface of the c-MYC:MAX complex, per-residue energetic decomposition with MM-GBSA to estimate how much each residue contributes to the binding free energy between MYC and MAX, and persistent salt bridge analysis to identify long-lived electrostatic interactions that stabilize the MYC-MAX interface. Residues detected by at least two of the three methods were taken as the strongest candidates for inhibitor design.

4.1 Trajectory-based pocket detection

Cavity detection was performed using MDpocket,²³ which extends the Voronoi tessellation-based Fpocket algorithm⁴⁵ to conformational ensembles. Rather than selecting a single representative frame for initial pocket identification, the three 600 ns production replicas were first superimposed onto a common crystal structure and concatenated into a single 1800 ns pooled trajectory prior to cavity detection.²³ MDpocket was run once on the pooled

trajectory, producing a single frequency map thresholded at the default isovalue (0.5).²³ Contiguous voxels above threshold (26-connectivity) were partitioned into individual sub-pockets; clusters smaller than 5 Å³ were discarded as noise. Residues with a C-alpha atom within 5 Å of a sub-pocket’s voxel envelope were assigned as that sub-pocket’s core lining residues.⁴⁶

This procedure identified six sub-pockets at the MYC-MAX interface (Figure 6, Table 1). Each sub-pocket was then re-analysed individually using MDpocket’s pocket-characterization mode. Results showed two sub-pockets (SP2 and SP3) containing lining residues from both MYC and MAX and were therefore interfacial. The remaining four (SP1, SP4-SP6) were single-chain, consistent with PPI interfaces distributing druggable contacts across several small hot-spot pockets rather than one large cavity.⁴³

Table 1: Consensus sub-pockets identified by MDpocket frequency grid clustering of the pooled 1800 ns combined trajectory (isovalue 0.5, minimum cluster size 5 Å³).

Sub-pocket	Frequency (mean/max)	Core lining residues
SP1	0.61 / 0.78	PRO230, LYS245, GLU248, TYR249 (MAX)
SP2	0.62 / 0.74	SER238, ARG239, ALA240 (MAX); ARG913, ARG914, LEU917 (MYC)
SP3	0.54 / 0.60	ARG214, ILE218 (MAX); PRO938, LYS939, VAL940 (MYC)
SP4	0.54 / 0.60	LYS945, ALA948, TYR949 (MYC)
SP5	0.52 / 0.54	LYS918, ARG914 (MYC)
SP6	0.52 / 0.54	ARG254, HIS258 , ASN257 (MAX)

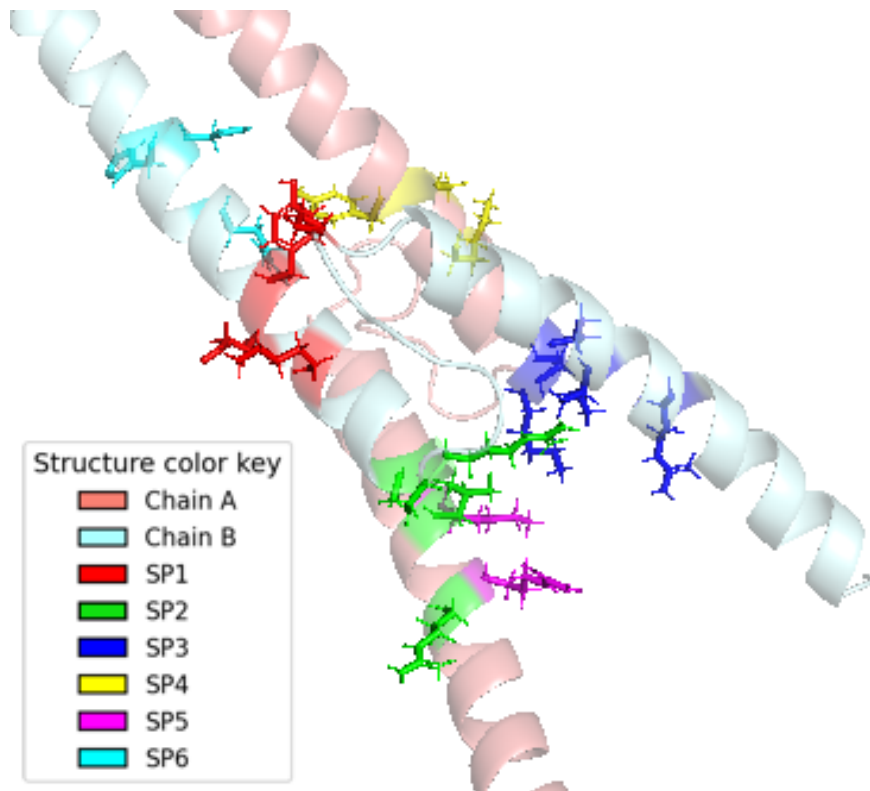


Figure 5: View of the three-dimensional localization of the 6 consensus sub-pockets at the MYC-MAX bHLHLZ interface. MYC (Chain A) is shown as a light red ribbon and MAX (Chain B) as a light blue ribbon.

Per-snapshot pocket volume, hydrophobicity score, local hydrophobic density, and polarity score were extracted for each cavity across the pooled 1800 ns trajectory using MD-pocket’s pocket-characterization mode. Pocket volume is the volume of the cavity identified in each snapshot and therefore measures its size. Hydrophobicity score represents the mean hydrophobicity of the residues lining the pocket, with higher values indicating a more hydrophobic pocket environment. Local hydrophobic density measures the density of hydrophobic residues or hydrophobic character within the local pocket environment, providing an indication of how concentrated hydrophobic regions are around the cavity. Polarity score represents the mean polarity of the residues lining the pocket, with higher values indicating a more polar pocket environment. Together, these descriptors characterize the cavity in terms of its size and the chemical properties of its lining, allowing the six candidate sub-pockets to be compared according to both their structural dimensions and physicochemical

environment.

	Pocket Volume (Å ³)	Hydrophobicity Score	Mean Local Hydrophobic Density	Polarity Score
SP1	193.07	11.60	28.27	3.62
SP2	174.60	8.96	16.74	3.48
SP3	126.47	13.93	14.78	1.59
SP4	103.30	0.22	1.89	3.19
SP5	84.57	7.28	3.94	2.23
SP6	89.82	28.35	5.79	1.62

Figure 6: Table of physicochemical properties of the MYC-MAX interface pocket across three replicas, as computed by MDpocket. Panels show Mean Pocket Volume (Å³), Hydrophobicity Score, Mean Local Hydrophobic Density and Polarity Score .

In figure 6, SP1 displayed the largest mean volume (193.07 Å³), followed by SP2 (174.6 Å³) and SP3 (126.47 Å³), while SP4–SP6 had mean volumes between 89.82 and 103.3 Å³. SP1 also showed the highest local hydrophobic density (28.27), together with a hydrophobicity score of 11.60 and the highest polarity score (3.62). SP2 presented a similar overall volume (174.6 Å³), with intermediate hydrophobicity (8.96), local hydrophobic density (16.74), and polarity (3.48). SP3 had a mean volume of 126.47 Å³ and a hydrophobicity score of 13.93, with a lower polarity score of 1.59. SP4 was characterized by the lowest hydrophobicity score (0.22) and local hydrophobic density (1.89), alongside a polarity score of 3.19. SP5 had a mean volume of 84.57 Å³, with moderate hydrophobicity (7.28), low local hydrophobic density (3.94), and a polarity score of 2.23. SP6 had a mean volume of 89.82 Å³ and the highest hydrophobicity score of the six pockets (28.35), with a local hydrophobic density of 5.79 and a polarity score of 1.62. Overall, SP1 and SP2 were characterized by larger cavity volumes and substantial hydrophobic and polar components, while SP6 exhibited a particularly high overall hydrophobicity within a smaller cavity.

4.2 Per-residue MM-GBSA decomposition

To identify energetically important residues independently of geometric criteria, per-residue MM-GBSA binding free energy decomposition was performed using `gmx_MMPBSA`.^{37,47} As is well-established for MM-GBSA calculations, absolute binding free energies are subject to systematic overestimation,²⁴ consequently the per-residue contributions reported here are used exclusively for relative ranking and not interpreted as quantitative thermodynamic values.³⁷

The residues with the most favourable decomposed contributions on the MYC side were, in descending order of contribution: ARG970, ARG925, ARG968, ARG913, ARG971, ARG919, ARG914, ASP965, ASP926, GLU916, GLU972 and GLU963 (Figure 7). On the MAX side, the dominant contributors were ARG254, ARG279, ARG239, ASP265, ASP244, ASP263, GLU248 and GLU275. The hotspot list is dominated by charged ARG, GLU and ASP residues on both chains, which is consistent with the electrostatic complementarity characterising bHLHLZ heterodimer interfaces and corroborates the salt bridge pattern described in Section 3.

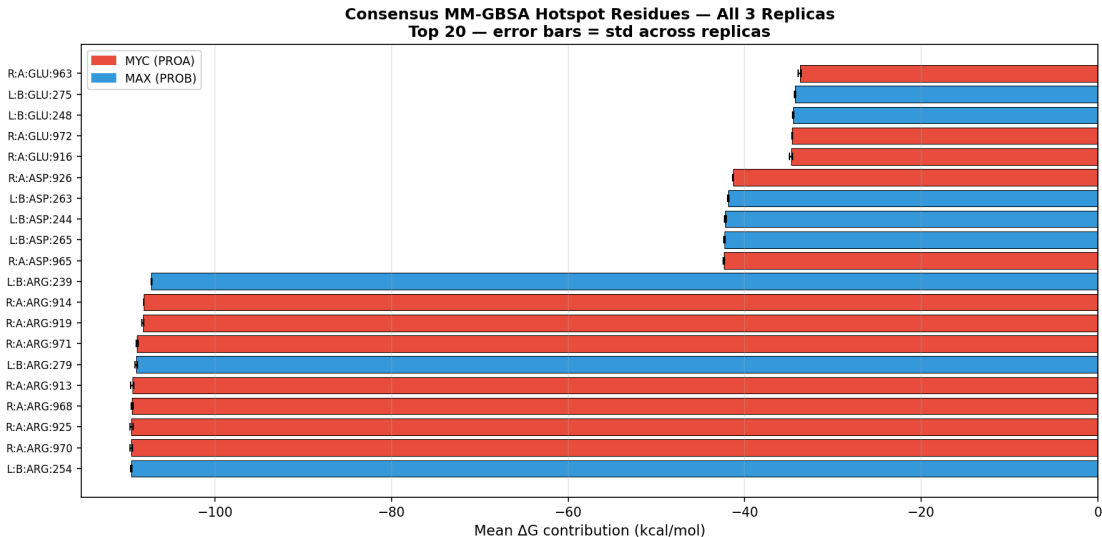


Figure 7: Per-residue MM-GBSA binding free energy decomposition for MYC-MAX interface residues, ranked by energetic contribution. Results represent the consensus across three independent replicas. Absolute values are systematically overestimated and should be interpreted as a relative ranking only.²⁴

4.3 Triple-method validation

Residues identified by two or more of the three methods of persistent salt bridge participation (Figure 4), MDpocket sub-pocket membership (Table 1), and MM-GBSA top 20 ranking (Figure 7) were taken as high-confidence interface hotspots (Figure 8 and Figure 9). No residue was simultaneously identified by all three methods, which is expected given that each method operates on a distinct physical observable: geometric cavity occupancy, energetic contribution and electrostatic persistence are not equivalent criteria and, in general, they will not select identical residues.

Cross-validation: Salt Bridges | MM-GBSA | MDpocket

Protein	Residue	Number	Salt Bridge	MM-GBSA	MDpocket	Score
MYC	ARG	913	-	✓	SP2	2/3
MYC	ARG	914	-	✓	SP2+SP5	2/3
MYC	GLU	916	-	✓	SP2+	2/3
MYC	ARG	970	✓	✓	-	2/3
MAX	ARG	239	-	✓	SP2	2/3
MAX	GLU	248	-	✓	SP1	2/3
MAX	ARG	254	-	✓	SP6	2/3
MAX	GLU	275	✓	✓	-	2/3



Figure 8: Multi-method validation pipeline of hotspot residues at the MYC-MAX interface.

Eight residues scored 2/3 across the three methods (Figure 8). On the MYC side, ARG913 and GLU916 were identified as both MM-GBSA hotspots and SP2 pocket-lining residues (916GLU with an extended cutoff of 8 Å). Although GLU916 narrowly exceeded the 5 Å static core cutoff in the reference structure (5.35 Å), frame-by-frame analysis of its nearest heavy atom across the pooled 1800 ns trajectory showed persistent proximity to SP2 (mean distance 3.50 Å), falling within the same 5 Å threshold in 90.7% of sampled

frames. ARG914 was also identified by MM-GBSA and it is a core residue on both SP2 (mean distance 3.32 Å) and SP5 (mean distance 4.42Å). ARG970 was classified as a MM-GBSA hotspot and a persistent salt bridge partner (ARG970-GLU275, ~70% occupancy). On the MAX side, GLU248, ARG239 and ARG254 were identified as MM-GBSA hotspots and corresponded to the SP1, SP2 and SP6 pocket-lining residues, respectively. Finally, GLU275 was identified as both MM-GBSA and a persistent salt bridge that corresponded to the ARG970 ~70% occupancy salt bridge mentioned before.

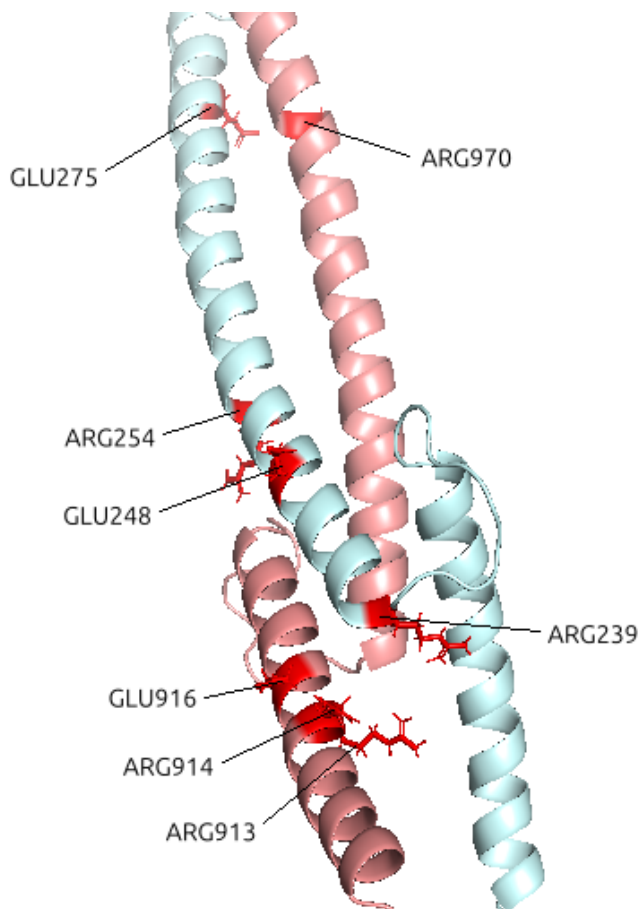


Figure 9: Structural localisation of multi-method validated hotspot residues at the MYC-MAX interface. Residues scoring 2/3 across the three methods are shown in red using Licorice representation. MYC is shown as a light red ribbon and MAX as a light blue ribbon.

Sub-pocket SP2 emerges as the most convergently validated cavity: four of its six constituent residues (ARG913, ARG914, GLU916, ARG239) independently scored 2/3 in the three-method consensus analysis, and SP2 spans both the MYC and MAX helices, mak-

ing it an interfacial rather than monomer-surface feature. The mixed electrostatic (ARG, GLU) and hydrophobic (ALA240, LEU917) composition of SP2 suggests it could accommodate inhibitors exploiting both polar and van der Waals contacts, a feature generally considered favourable for the design of PPI disruptors.⁴³ Taken together, the convergent geometric, energetic and structural evidence positions SP2 as the primary candidate cavity for structure-based inhibitor design targeting the MYC-MAX interface.

SP1 also emerges as a promising candidate based on its pocket characteristics. It displayed the highest mean pocket volume, local hydrophobic density, and polarity score among the six characterized cavities. Although SP1 is associated with a single chain rather than the MYC-MAX interface, one of its four identified residues (GLU248) scored 2/3 in the cross-validation analysis, providing additional support for the structural relevance of this cavity. Thus, SP1 represents a complementary candidate to SP2, with its prioritization supported primarily by its pocket geometry and physicochemical characteristics, whereas SP2 benefits from both these features and its direct localization at the MYC-MAX interface.

Notably, both partners of the ARG970(MYC)-GLU275(MAX) salt bridge (Section 3, ~70% occupancy) independently scored 2/3 in the cross-validation analysis, being identified as MM-GBSA hotspots on both the MYC and MAX sides despite lying outside any of the six geometrically-defined sub-pockets. This bidirectional convergence – both interacting partners, rather than a single residue, being flagged by an independent energetic method – reinforces the role of this salt bridge as a persistent electrostatic anchor at the MYC-MAX interface, complementing the hydrophobic and mixed-character sub-pockets discussed above.

Conclusion

c-MYC is a central oncogenic transcription factor whose intrinsic structural disorder has long precluded conventional structure-based drug design. In this work we have performed MD simulations of two systems: isolated c-MYC and the complex c-MYC:MAX in a physiological

NaCl solution (0.15 M) across three independent replicas of each system (total simulation time of 3.6 μ s), MAX binding markedly stabilized the c-MYC structure, as evidenced by a marked reduction in the RMSF profile and a decrease in the average backbone RMSD from 10.55 Å for isolated c-MYC to 2.66 Å in the c-MYC complex. This stabilization was further reflected by reduced per-residue flexibility and an increased radius of gyration, consistent with a coupled folding-and-binding mechanism.⁴² The resulting MYC-MAX interface remained highly stable, sustained by a mean of 437 persistent atomic contacts and anchored by two main salt bridges, LYS256(MAX)-GLU957(MYC) and ARG970(MYC)-GLU275(MAX).

Given the intrinsically disordered and highly dynamic nature of isolated c-MYC in solution observed in this work, the disruption of the MYC-MAX protein-protein interaction offers a validated alternative strategy, so a residue-level understanding of the binding interface is a necessary first step toward rational inhibitor design. To this end, we applied a three-method validation pipeline combining MDpocket cavity detection, MM-GBSA decomposition, and salt bridge persistence analysis, which together identified six main sub-pockets at the interface. Among these, SP2 emerged as the most consistently validated cavity: three of its six constituent residues (ARG913, ARG914, and ARG239) scored 2/3 across the three independent methods, along with a fourth residue, GLU916, validated separately using an extended distance cutoff. These results provide concrete candidate sites for inhibitors aimed at disrupting MYC-MAX dimerization.

However, some limitations should be noted. First, the absolute binding energies estimated by MM-GBSA are known to be systematically overestimated and were therefore used exclusively for relative ranking. Secondly, the simulations were initiated from the DNA-bound crystal structure with the DNA removed, and the influence of DNA on the geometry of the interfacial pocket was not evaluated. These structural results are intended to serve as the foundation for future work, including conformational sampling of isolated c-MYC and machine learning screening of candidate compounds against the identified interface hotspots.

- (7) Hermeking, H. The MYC oncogene as a cancer drug target. *Current cancer drug targets* **2003**, *3*, 163–175.
- (8) Jakobsen, S. T.; Siersbæk, R. Transcriptional regulation by MYC: an emerging new model. *Oncogene* **2024**, 1–7.
- (9) Blackwood, E. M.; Eisenman, R. N. Max: a helix-loop-helix zipper protein that forms a sequence-specific DNA-binding complex with Myc. *Science* **1991**, *251*, 1211–1217.
- (10) Atchley, W. R.; Fitch, W. M. Myc and Max: molecular evolution of a family of proto-oncogene products and their dimerization partner. *Proceedings of the National Academy of Sciences* **1995**, *92*, 10217–10221.
- (11) Lourenco, C.; Resetca, D.; Redel, C.; Lin, P.; MacDonald, A. S.; Ciaccio, R.; Kenney, T. M.; Wei, Y.; Andrews, D. W.; Sunnerhagen, M.; others MYC protein interactors in gene transcription and cancer. *Nature Reviews Cancer* **2021**, *21*, 579–591.
- (12) Ge, Y.; Lu, H.; Marti, J. Conformational drugging pockets at the surface of c-MYC proto-oncogenes. *bioRxiv* **2026**, 2026–05.
- (13) Lu, H.; Marti, J.; Faraudo, J. Probing the Structural Dynamics of the Unbound MAX Protein: Insights from Well-Tempered Metadynamics. *Journal of chemical information and modeling* **2025**,
- (14) Ross, J.; Miron, C. E.; Plescia, J.; Laplante, P.; McBride, K.; Moitessier, N.; Möröy, T. Targeting MYC: From understanding its biology to drug discovery. *European Journal of Medicinal Chemistry* **2021**, *213*, 113137.
- (15) Llombart, V.; Mansour, M. R. Therapeutic targeting of “undruggable” MYC. *EBioMedicine* **2022**, *75*.
- (16) Li, Z.; Huang, Y.; Hung, T. I.; Sun, J.; Aispuro, D.; Chen, B.; Guevara, N.; Ji, F.; Cong, X.; Zhu, L.; others MYC-targeting inhibitors generated from a stereodiversified

- bicyclic peptide library. *Journal of the American Chemical Society* **2024**, *146*, 1356–1363.
- (17) Whitfield, J. R.; Soucek, L. MYC in cancer: from undruggable target to clinical trials. *Nature Reviews Drug Discovery* **2025**, 1–13.
- (18) Kiessling, A.; Sperl, B.; Hollis, A.; Eick, D.; Berg, T. Selective inhibition of c-Myc/Max dimerization and DNA binding by small molecules. *Chemistry & biology* **2006**, *13*, 745–751.
- (19) Wang, C.; Zhang, J.; Yin, J.; Gan, Y.; Xu, S.; Gu, Y.; Huang, W. Alternative approaches to target Myc for cancer treatment. *Signal transduction and targeted therapy* **2021**, *6*, 117.
- (20) Clausen, D. M.; Guo, J.; Parise, R. A.; Beumer, J. H.; Egorin, M. J.; Lazo, J. S.; Prochownik, E. V.; Eiseman, J. L. In vitro cytotoxicity and in vivo efficacy, pharmacokinetics, and metabolism of 10074-G5, a novel small-molecule inhibitor of c-Myc/Max dimerization. *The Journal of pharmacology and experimental therapeutics* **2010**, *335*, 715–727.
- (21) Huang, M.-J.; Cheng, Y.-c.; Liu, C.-R.; Lin, S.; Liu, H. E. A small-molecule c-Myc inhibitor, 10058-F4, induces cell-cycle arrest, apoptosis, and myeloid differentiation of human acute myeloid leukemia. *Experimental hematology* **2006**, *34*, 1480–1489.
- (22) Soucek, L.; Jucker, R.; Panacchia, L.; Ricordy, R.; Tato, F.; Nasi, S. Omomyc, a potential Myc dominant negative, enhances Myc-induced apoptosis. *Cancer research* **2002**, *62*, 3507–3510.
- (23) Schmidtke, P.; Bidon-Chanal, A.; Luque, F. J.; Barril, X. MDpocket: open-source cavity detection and characterization on molecular dynamics trajectories. *Bioinformatics* **2011**, *27*, 3276–3285.

- (24) Genheden, S.; Ryde, U. The MM/PBSA and MM/GBSA methods to estimate ligand-binding affinities. *Expert opinion on drug discovery* **2015**, *10*, 449–461.
- (25) Deshpande, N.; Address, K. J.; Bluhm, W. F.; Merino-Ott, J. C.; Townsend-Merino, W.; Zhang, Q.; Knezevich, C.; Xie, L.; Chen, L.; Feng, Z.; Kramer Green, R.; Flippen-Anderson, J. L.; Westbrook, J.; Berman, H. M.; Bourne, P. E. The RCSB Protein Data Bank: a redesigned query system and relational database based on the mmCIF schema. *Nucleic Acids Research* **2005**, *33*, D233–D237.
- (26) Kouranov, A.; Xie, L.; de la Cruz, J.; Chen, L.; Westbrook, J.; Bourne, P. E.; Berman, H. M. The RCSB PDB information portal for structural genomics. *Nucleic acids research* **2006**, *34*, D302–D305.
- (27) Jorgensen, W. L.; Chandrasekhar, J.; Madura, J. D.; Impey, R. W.; Klein, M. L. Comparison of simple potential functions for simulating liquid water. *The Journal of chemical physics* **1983**, *79*, 926–935.
- (28) Jo, S.; Kim, T.; Iyer, V. G.; Im, W. CHARMM-GUI: a web-based graphical user interface for CHARMM. *Journal of computational chemistry* **2008**, *29*, 1859–1865.
- (29) Brooks, B. R.; Brooks III, C. L.; Mackerell Jr, A. D.; Nilsson, L.; Petrella, R. J.; Roux, B.; Won, Y.; Archontis, G.; Bartels, C.; Boresch, S.; others CHARMM: the biomolecular simulation program. *Journal of computational chemistry* **2009**, *30*, 1545–1614.
- (30) Lee, J.; Cheng, X.; Jo, S.; MacKerell, A. D.; Klauda, J. B.; Im, W. CHARMM-GUI input generator for NAMD, GROMACS, AMBER, OpenMM, and CHARMM/OpenMM simulations using the CHARMM36 additive force field. *Biophysical journal* **2016**, *110*, 641a.
- (31) Huang, J.; Rauscher, S.; Nawrocki, G.; Feig, M. CHARMM36m: an improved force field for folded and intrinsically disordered proteins. *Nature Methods* **2017**, *14*, 71–73.

- (32) Hess, B.; Bekker, H.; Berendsen, H. J. C.; Fraaije, J. G. E. M. LINCS: A linear constraint solver for molecular simulations. *Journal of Computational Chemistry* **1997**, *18*, 1463–1472.
- (33) Berendsen, H. J.; van der Spoel, D.; van Drunen, R. GROMACS: A message-passing parallel molecular dynamics implementation. *Computer physics communications* **1995**, *91*, 43–56.
- (34) Essmann, U.; Perera, L.; Berkowitz, M. L.; Darden, T.; Lee, H.; Pedersen, L. G. A smooth particle mesh Ewald method. *The Journal of Chemical Physics* **1995**, *103*, 8577–8593.
- (35) Bussi, G.; Donadio, D.; Parrinello, M. Canonical sampling through velocity rescaling. *The Journal of Chemical Physics* **2007**, *126*, 014101.
- (36) Bernetti, M.; Bussi, G. Pressure control using stochastic cell rescaling. *The Journal of Chemical Physics* **2020**, *153*, 114107.
- (37) Valdés-Tresanco, M. S.; Valdés-Tresanco, M. E.; Valiente, P. A.; Moreno, E. gmx_MMPBSA: A New Tool to Perform End-State Free Energy Calculations with GROMACS. *Journal of Chemical Theory and Computation* **2021**, *17*, 6281–6291.
- (38) Harris, C. R. et al. Array programming with NumPy. *Nature* **2020**, *585*, 357–362, Version 1.26.4.
- (39) Virtanen, P. et al. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods* **2020**, *17*, 261–272, Version 1.14.1.
- (40) Hunter, J. D. Matplotlib: A 2D graphics environment. *Computing in Science & Engineering* **2007**, *9*, 90–95, Version 3.7.3.
- (41) Humphrey, W.; Dalke, A.; Schulten, K. VMD: visual molecular dynamics. *Journal of molecular graphics* **1996**, *14*, 33–38.

- (42) Dyson, H. J.; Wright, P. E. Intrinsically unstructured proteins and their functions. *Nature Reviews Molecular Cell Biology* **2005**, *6*, 197–208.
- (43) Wells, J. A.; McClendon, C. L. Reaching for High-Hanging Fruit in Drug Discovery at Protein-Protein Interfaces. *Nature* **2007**, *450*, 1001–1009.
- (44) Ran, X.; Gestwicki, J. E. Inhibitors of protein–protein interactions (PPIs): an analysis of scaffold choices and buried surface area. *Current Opinion in Chemical Biology* **2018**, *44*, 75–86, Next Generation Therapeutics.
- (45) Le Guilloux, V.; Schmidtke, P.; Tuffery, P. Fpocket: An open source platform for ligand pocket detection. *BMC Bioinformatics* **2009**, *10*, 168.
- (46) Gao, M.; Skolnick, J. The distribution of ligand-binding pockets around protein–protein interfaces suggests a general mechanism for pocket formation. *Proceedings of the National Academy of Sciences* **2012**, *109*, 3784–3789.
- (47) Miller, B. R.; McGee, T. D.; Swails, J. M.; Homeyer, N.; Gohlke, H.; Roitberg, A. E. MMPBSA.py: An Efficient Program for End-State Free Energy Calculations. *Journal of Chemical Theory and Computation* **2012**, *8*, 3314–3321.